Venkata Lakshmi Parimala Pasupuleti

Sr. AI Product Engineer · Prototype to Production · LLM APIs · Agents · Python · AWS · Salesforce · Snowflake · System Integrations

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SUMMARY

Senior software engineer with 9+ years of production engineering depth and a pattern of taking AI prototypes from assessment to deployed systems that real users depend on. Built **Leasel** from scratch: reviewed the architecture options, documented requirements, built the data models, APIs, integrations, error handling, tests, and deployment processes - and operate it in production with a CI regression gate and monitoring. Built **AgentProbe**, an open-source LLM evaluation toolkit with automated testing and regression harness built specifically to catch what AI-generated code gets wrong - exactly the "careful review of generated code" Cardiff requires. At Entrata (9+ years, 100K+ users), owned enterprise system integrations, data pipeline reliability, and production incident resolution end-to-end across a regulated multi-tenant platform. Strong Python, TypeScript/JavaScript, SQL, REST APIs, AWS, Salesforce, Snowflake, Docker, and CI/CD. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

Prototype Assessment & Production Development · LLM APIs · AI Agents · Evaluations · Python · TypeScript · JavaScript · REST APIs · System Integrations · AWS (S3 · EC2 · RDS · Lambda) · Salesforce · Snowflake · SQL · PostgreSQL · Data Modeling · Release Controls · CI/CD · GitHub Actions · Production Monitoring · Incident Response · LangGraph · LangChain · RAG Pipelines · Automated Testing · Regression Harness · MCP · Enterprise Integrations · Docker · Kubernetes · Stakeholder Communication · Regulated & Compliant Systems · FastAPI · Django

EXPERIENCE

AI/ML Engineer (Volunteer)

Sep 2026 - Present

Blue Horse Digital — Virginia, USA

- Building ML models, AI-powered features, and data pipelines end-to-end; connecting AI components to backend APIs and surfacing insights into reporting layers for business stakeholders.

AI/ML Engineer Intern

May 2025 - Aug 2025

Magizhs Tech Solution

- Assessed existing AI prototype, documented production requirements, rebuilt with proper data models, error handling, and validation guardrails - delivered as a production LangGraph agentic system with FastAPI serving and CI/CD pipeline.
- **Deployed MCP server** connecting agent outputs to downstream enterprise systems - defined the integration contract, handled authentication, built failure routing, and enforced access boundaries; standardized inconsistent data inputs into clean structured outputs across the integration boundary.

Senior Software Engineer

Sep 2018 - Aug 2024

Entrata - PropTech / Real Estate SaaS (100K+ monthly active users, multi-tenant)

- **Owned enterprise system integrations and REST API contracts** across a regulated multi-tenant platform - third-party integrations with defined input/output contracts, data normalization from inconsistent external sources, OAuth flows, rate-limit handling, retry semantics, and long-term production maintenance.
- **Built and operated production data pipelines end-to-end** - annual HUD compliance pipeline (federal rows → ~180,000 normalized domain records) powering eligibility and rent calculations for 100K+ housing units; 7 years of zero violations; PHI-encrypted AWS S3 migration (100% coverage, zero security incidents).
- **Owned production incident response end-to-end** - built internal debugging tools reducing debugging time by 40%; led root-cause analysis and postmortems; drove corrective actions and documented follow-up work; maintained bug count below 10 for 6+ consecutive months.
- Established release controls and CI/CD automation (Python/Bash pipelines) eliminating 80% of manual operational workload; optimized 200+ SQL queries (30% gain); shipped 20+ full-stack production modules across 120+ Agile sprints; mentored 5+ engineers.

Associate Software Engineer

Jul 2017 - Sep 2018

Entrata India, Pune

- Built backend modules (Python, PHP, Java, PostgreSQL); led AWS S3 cloud migration; delivered large-scale data import workflows (1.6M records) with data validation and reconciliation.

PROJECTS

Leasel - AI Prototype to Production (Assess → Build → Ship → Own) 2026 | Python, LangGraph (8-agent), FastAPI, SQLite, CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

- Took the problem from prototype assessment to production: documented architecture options, built data models, APIs, integrations, error handling, tests, and CI/CD deployment process - then shipped three production surfaces sharing one SQLite knowledge repository and operate them with a regression gate and monitoring. The same prototype-to-production ownership Cardiff needs for its lender routing, decisioning workflows, and sales calculator agents.
- 93.6% precision and 87.9% F1 across 1,240 test cases; CI regression harness catches regressions before they reach users.

AgentProbe - LLM Evaluation & Release Controls Toolkit 2025 | Python, FastAPI, LangGraph, CI/CD | github.com/venkatalakshmiparimala/AgentProbe

- Open-source LLM evaluation toolkit built to enforce careful review of AI-generated outputs before they reach production: CI-integrated regression harness, automated testing with golden datasets, LLM-as-judge scoring, and hallucination detection across Anthropic, OpenAI, Azure OpenAI, Groq, and Ollama. Used daily as a release control layer for AI systems.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia I GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017